

TENNIS DATA ANALYSIS

Tennis Matches Analysis Report

A Data-Driven Exploration of Professional Tennis Match Data

PRESENTED BY

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24,079

APPEARANCES

16,873

MATCHES

2,644

UNIQUE PLAYERS



◆ 2024

01

SECTION ONE / THREE

Player Profiles & Fan Behavior

PRESENTED BY

Mehrnoosh Yaghubi

QUESTIONS IN THIS SECTION

Q1 Unique players & home/away labeling

Q10 Height vs ranking correlation

Q15 Distinct countries represented

OP2 Fan votes vs betting market

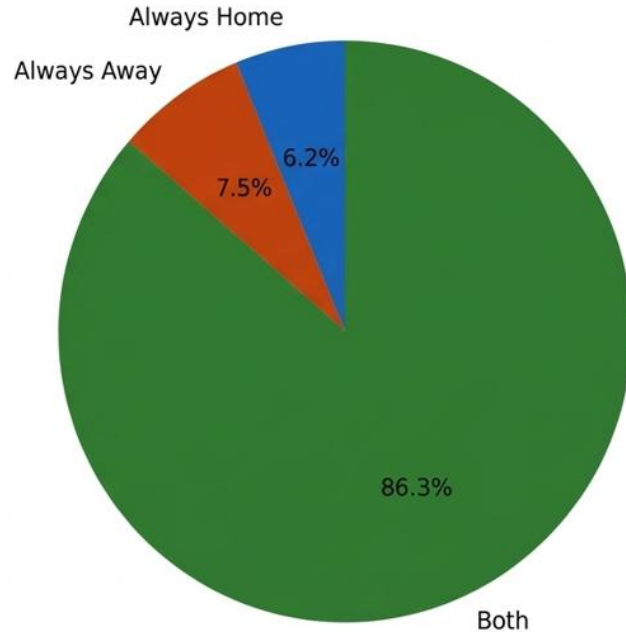
Q2 Average player height

Q13 Left vs right-handed players

OP1 Fan vote home bias

FIG. 1 · HOME/AWAY DISTRIBUTION

Chart 2: Unique players -- Always Home / Always Away / Both
(counted per player, not per row)



How Many Unique Tennis Players Are in the Dataset?

2,644 unique players were identified from 24,079 home/away appearances across 16,873 matches — an average of ~9.1 matches per player. A side investigation into whether the 'home/away' label carries meaning found that 86.3% of players appeared on both sides, while 13.7% were consistently assigned to one side.

2,644

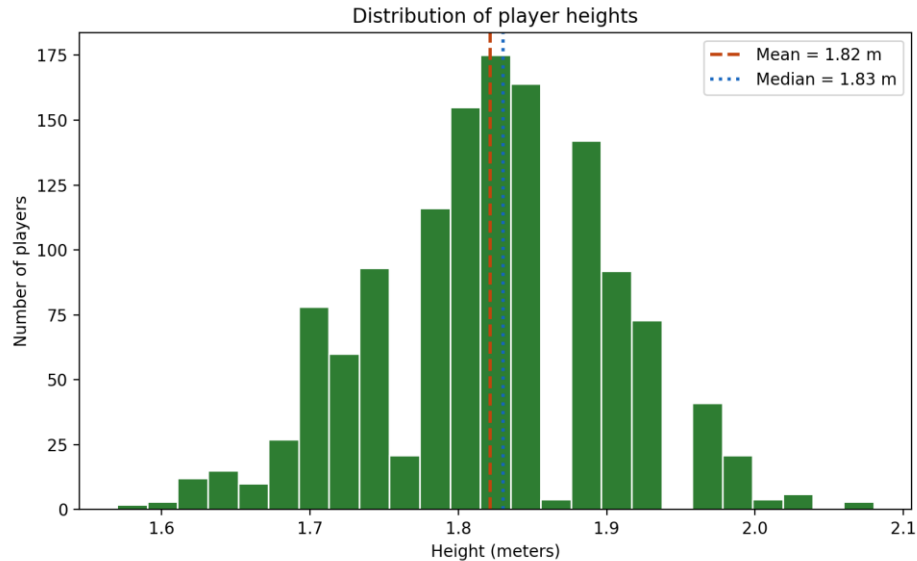
unique players

52 / 2,644

show systematic one-side bias (~2%)

METHODOLOGY: Row-level: 51.5% home vs. 48.5% away — no raw
The 52 players with ≥ 5 appearances always on one side are statistically significant, suggesting a source-data labeling rule (e.g., seeding), not an error.

FIG. 2 · Distribution of Player Heights



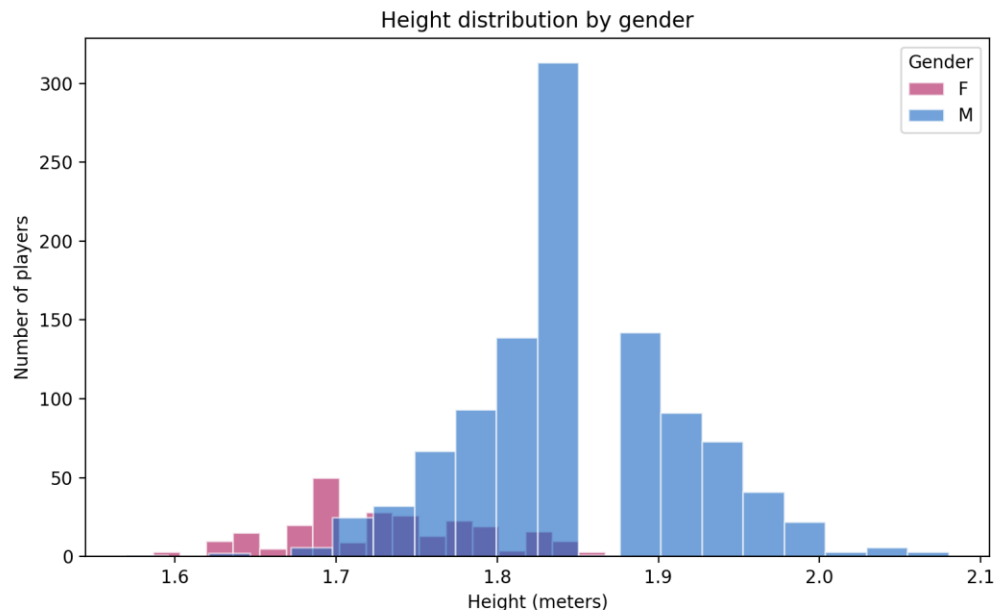
What Is the Average Height of the Players?

The overall mean player height is 1.82 m. The distribution is broad, reflecting the mixed-gender composition of the dataset. The wide spread indicates that male and female players contribute distinct height ranges, pulling the combined distribution across a large interval around the mean.

| 1.843 m

METHODOLOGY: Reporting a single overall average without noting the gender imbalance would be misleading.

FIG. 3 · HEIGHT HISTOGRAM BY GENDER



What Is the Average Height of the Players?

The height distribution is genuinely bimodal — a smaller cluster of women around 1.65–1.75 m and a larger cluster of men around 1.78–1.95 m. The ~11 cm gap between genders is consistent with known professional tennis averages (WTA ~1.70–1.75 m, ATP ~1.85 m). The overall mixed average of 1.82 m sits closer to the men's mean simply because men outnumber women ~4:1 in the height-available subset.

1.843 m

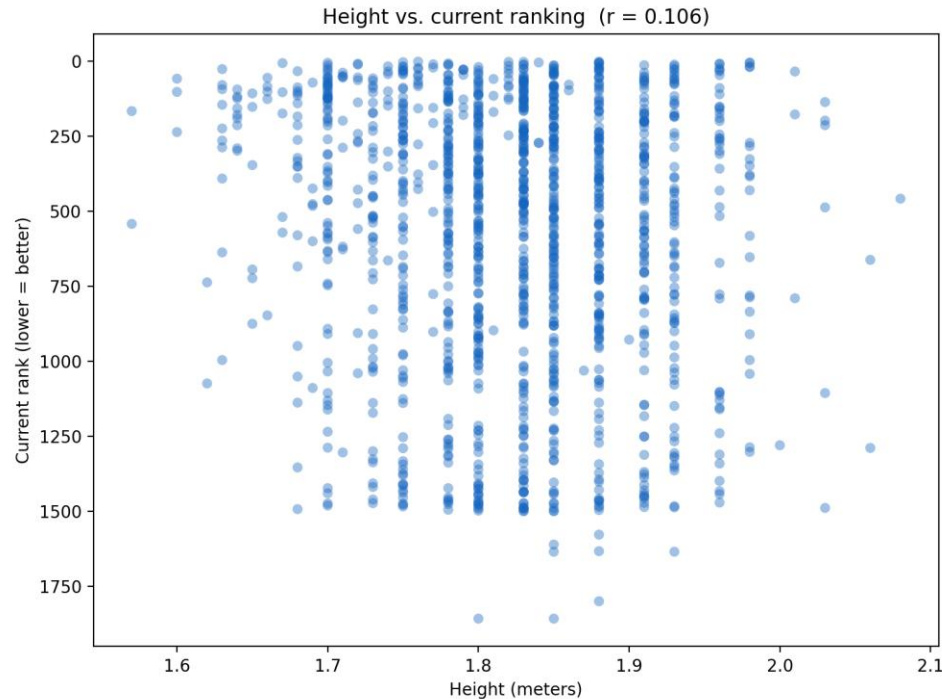
Men (n = 1,060)

1.732 m

Women (n = 256)

METHODOLOGY: Standard deviation is similar between genders (0.069 vs. 0.061).

FIG. 4 · HEIGHT VS CURRENT RANK



Is There a Correlation Between a Player's Height and Their Ranking?

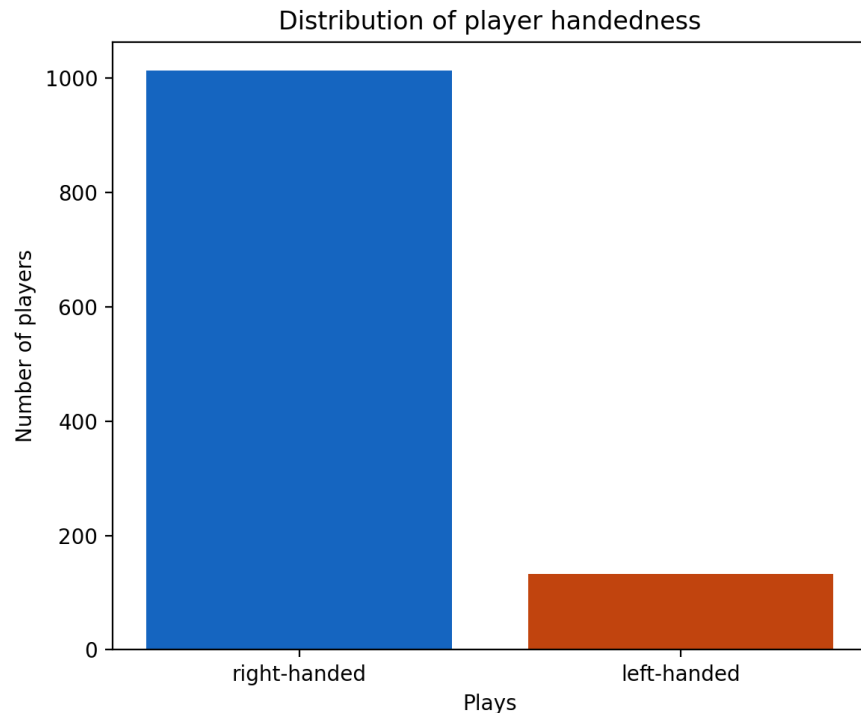
Pearson $r = 0.106$ indicates a positive but negligible linear relationship — $|r| < 0.2$ is generally considered insignificant. At every height value, players are spread across the full range of rankings rather than clustering toward better ranks for taller players. The slightly positive sign (taller $\rightarrow$ higher/worse rank) runs opposite to the common belief that height is an advantage, but the near-zero magnitude is likely noise.

$|r = 0.106$

negligible correlation
($|r| < 0.2 \rightarrow$ insignificant)

METHODOLOGY: Based on 1,307 players (49.4% of 2,644 — the rest are missing height or current_rank). Heights form distinct vertical bands (1.70, 1.73, 1.75, 1.78 m), suggesting original recording in feet/inches and conversion to rounded meters — a data characteristic, not an error.

FIG. 5 · HANDEDNESS DISTRIBUTION



What Is the Distribution of Left-Handed vs Right-Handed Players?

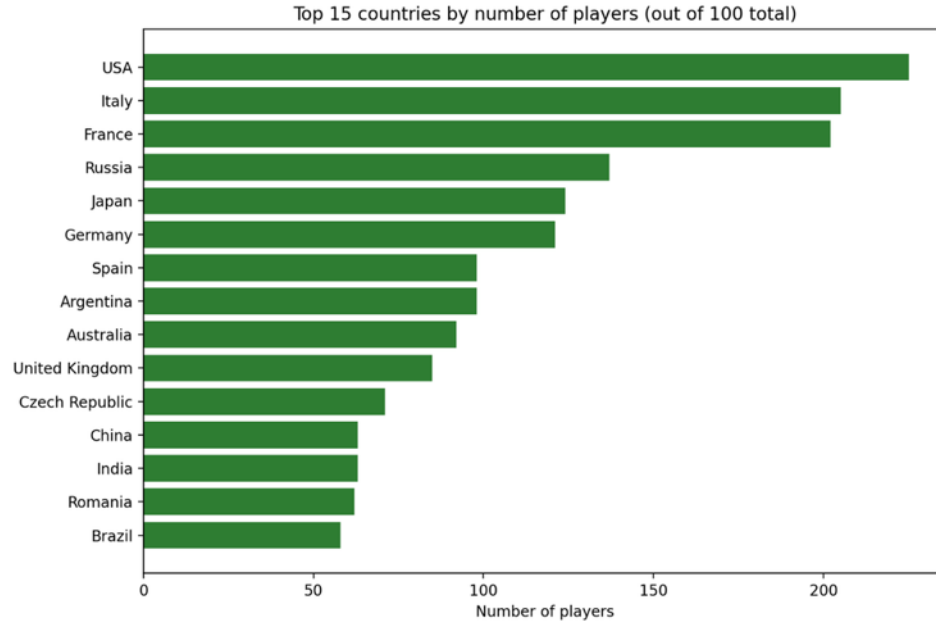
Among players with a recorded handedness, 88.4% are right-handed and 11.6% are left-handed. The left-handed rate is consistent with real-world patterns — roughly 10% of the general population is left-handed, and professional tennis is commonly observed to have a slightly higher share, as left-handed shots/spin are less familiar to right-handed-dominant opponents, offering a modest tactical edge.

1,013 · 88.4%
Right-handed

133 · 11.6%
Left-handed

METHODOLOGY: Based on only 1,146 of 2,644 unique players (43.3%) — the remaining 1,498 (56.7%) have no recorded 'plays' value. Should be reported as "among players with known handedness", not generalized to the full player population.

FIG. 6 · COUNTRIES REPRESENTED



How Many Distinct Countries Are Represented in the Dataset?

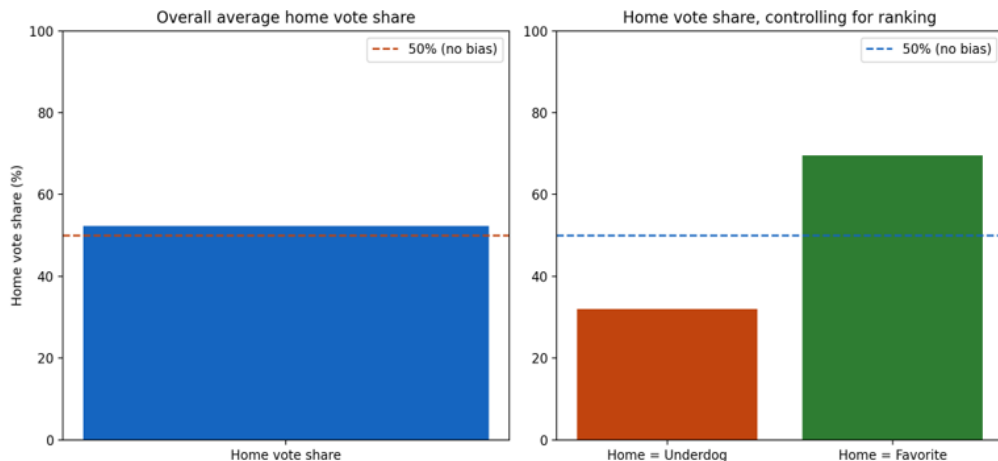
The dataset spans 100 distinct countries, reflecting the deeply global nature of professional tennis. Only 1 player has no recorded country, indicating near-complete geographic coverage. This breadth mirrors the international structure of the ATP, WTA, and ITF circuits, which span every continent and draw players from both traditional powerhouses and emerging tennis nations.

100

distinct countries represented
across 2,644 players

METHODOLOGY Country attribution from player records. Near-complete complete coverage (only 1 player missing country). The global spread is consistent with the international tournament structure across ATP, across ATP, WTA, and ITF circuits.

FIG. 7 · FAN VOTE SHARE BY MATCHUP



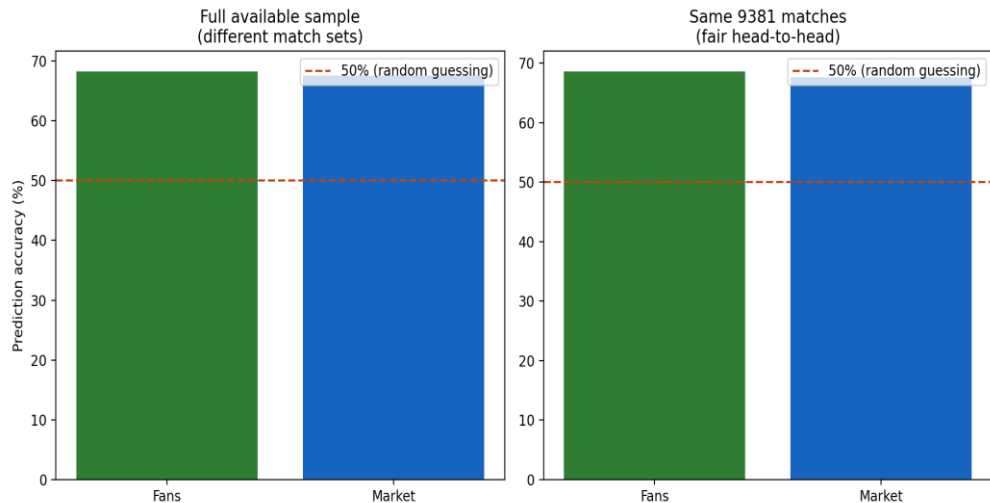
Is Fan Vote Share Systematically Biased Toward the 'Home' Label?

A Simpson's-paradox-like result. The overall 52.25% home vote share looks like a small home bias, but after controlling for ranking, fans correctly favor whichever player is actually better-ranked — regardless of the home/away label. When home is the underdog, fans shift to away (32% for home); when home is the favorite, fans favor home (69.5%). **The labeling-bias hypothesis is REJECTED.**

- | **52.25%** overall home vote share ($t = 8.52$, $p < 0.0001$)
- | **69.52%** when home is the favorite ($n = 5,206$)
- | **32.01%** when home is the underdog ($n = 4,427$)

METHODOLOGY The raw average sits slightly above 50% only because matches where home is better-ranked ($n=5,206$) slightly outnumber those where home is the underdog ($n=4,427$) — a sample composition effect, not psychological bias.

FIG. 8 · FAN VS MARKET ACCURACY



Fan Votes vs. Betting Market — Which Predicts the Winner More Accurately?

Both sources performed ~18 percentage points above the 50% random-guessing baseline, and the gap between them is only 0.9 percentage points — likely within statistical noise. This is counter-intuitive: betting markets are usually assumed to outperform crowd opinion because real money is at stake. A plausible explanation is that this dataset is dominated by lower-tier (ITF/Challenger) tournaments where neither side has deeper analysis.

68.6%

Fan accuracy (n = 9,381)

67.7%

Betting market accuracy (n = 9,381)

Δ 0.9 pp likely within statistical noise

METHODOLOGY Fair head-to-head on the same 9,381 matches where both sources had data. Full samples: fan 68.3% (n=13,475), market 67.6% (n=9,477). Tied/missing predictions excluded from accuracy calculations.

SECTION TWO / THREE

02 Match Statistics & Performance

PRESENTED BY Keyvan Hajizade

QUESTIONS IN THIS SECTION

Q7 Average aces per match

Q12 Games per set: men vs women

Q17 Average breaks of serve per match

OP2 Court surface effect on aces

Q8 Double faults by gender

Q16 Win rate vs top-10 opponents

OP1 Do winners hit more aces?

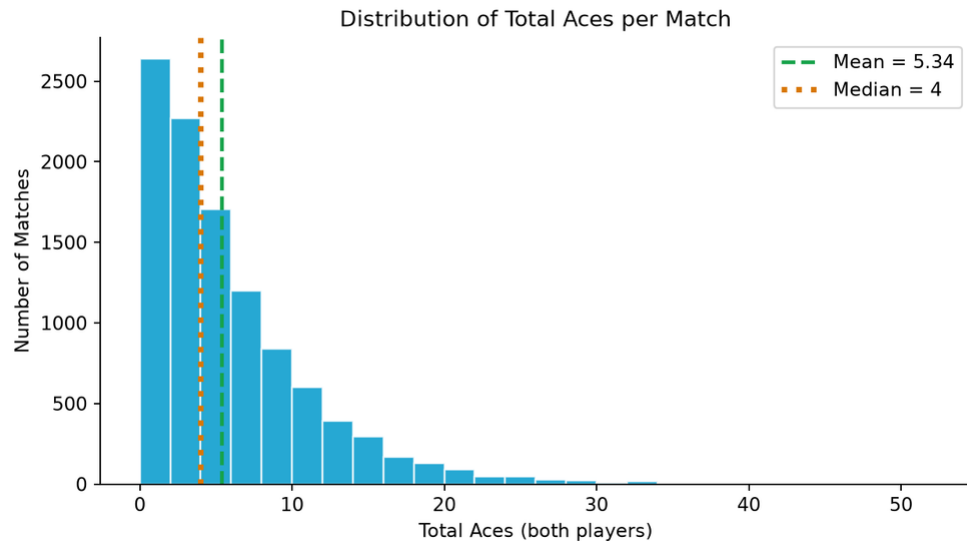


FIG.9 — DISTRIBUTION OF TOTAL ACES PER MATCH (N = FULL MATCH SAMPLE)

What Is the Average Number of Aces per Aces per Match?

The mean total aces per match is 5.34 while the median is 4 — this discrepancy indicates a right-skewed distribution, clearly visible in the chart. Approximately 50% of matches feature between 0 and 6 aces, suggesting that most matches are not serve-dominated. Outliers on the right side are genuine data points from long five-set matches and were retained.

5.34

Mean aces per match · Median: 4

Right skew caused by long five-set matches with elevated ace counts. Outliers were retained as genuine data rather than removed. The mean–median gap confirms the distribution is not normally distributed.

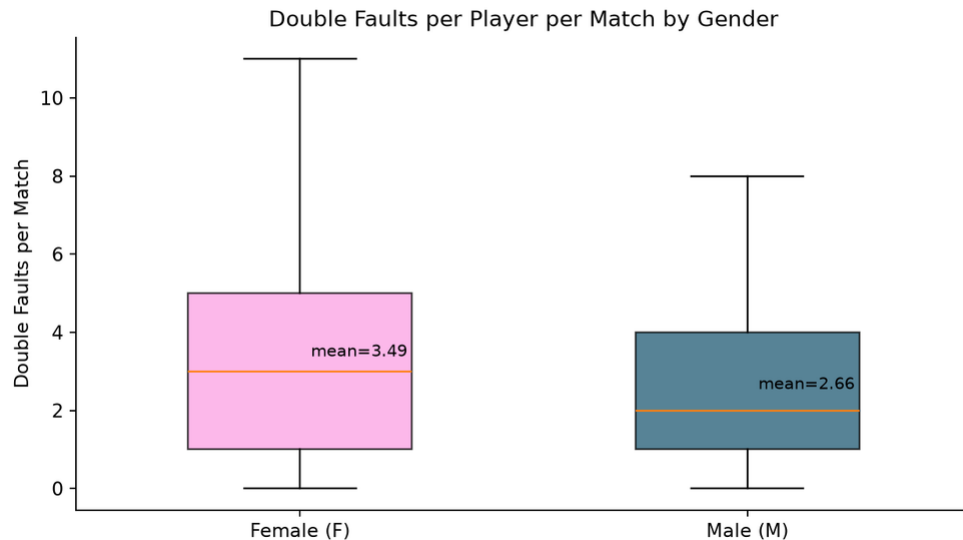


FIG. 10 — DOUBLE FAULTS PER MATCH, BY PLAYER GENDER

Is There a Difference in Double Faults Based on Gender?

Female players average 3.49 double faults per match, noticeably higher than the 2.66 recorded for men. The median values (3 vs 2) reflect the same pattern. The women's box plot shows a larger interquartile range, indicating greater dispersion in double fault counts among female players. Given the large sample size, the results are reliable.

3.49

Women · mean

2.66

Men · mean

Outliers excluded from the plot for readability. Larger IQR for women indicates greater variability. The gender gap is statistically significant given the sample size.

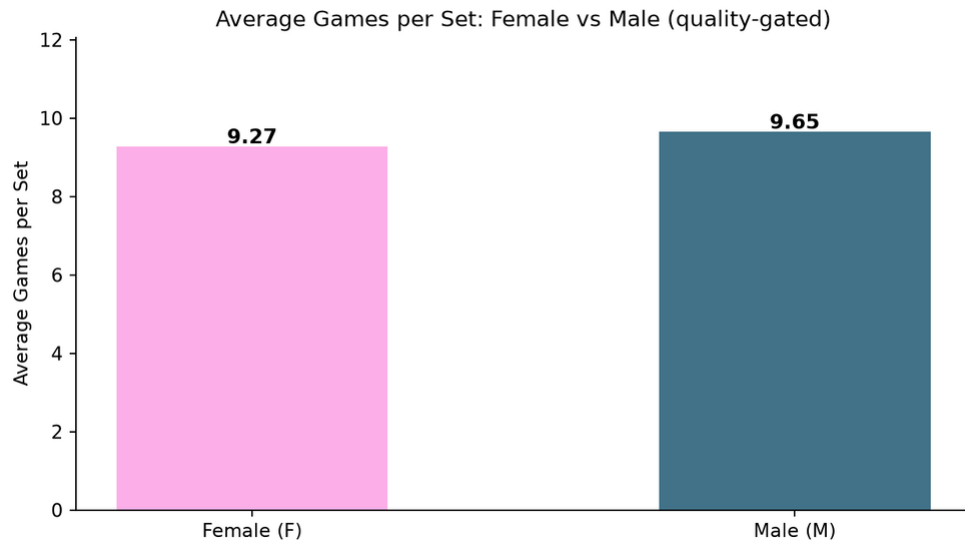


FIG. 11 — AVERAGE GAMES PER SET, MEN'S VS WOMEN'S MATCHES

Average Games per Set — Men's vs Women's Women's Matches?

The average number of games per set is 9.65 for men and 9.27 for women — a difference of approximately half a game. The median is 9 for both groups, meaning the discrepancy lies in longer sets and tie-breaks. This slight difference may stem from women playing best-of-three while men play best-of-five, or from the differing scoring structures in five-set matches.

9.65

Men · avg games / set

9.27

Women · avg games / set

617 incomplete sets (0-0 or 3-2 from retirements/walkovers) were excluded from the total of 20,461 sets to prevent artificial downward skew. Both groups have a median of 9 games per set.

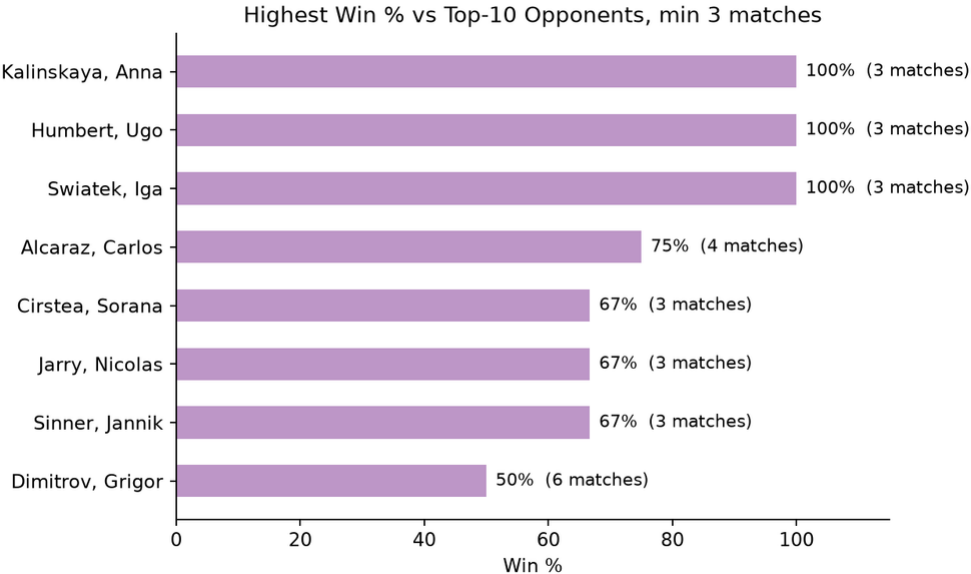


FIG. 12 — WINNING PERCENTAGE AGAINST TOP-10 OPPONENTS

Which Player Has the Highest Winning Percentage Against Top-10 Opponents?

The three players at the top of the table each played only three matches — too small a sample for statistical significance. The more reliable performances are Alcaraz (75% win rate over 4 matches) and Dimitrov (50% over 6 matches). However, the analysis uses the `current_rank` column, meaning the opponent's current ranking is used rather than the ranking they held at the time of the match.

75%

Alcaraz · 4 matches vs top-10

⚠ RELIABILITY FLAG · NOT STATISTICALLY SIGNIFICANT

This analysis is NOT reliable. The `current_rank` column reflects today's ranking, not the ranking at match time.

⚠ This analysis is NOT reliable. The `current_rank` column reflects today's ranking, not the ranking at match time. Three-match samples at the top are too small for statistical conclusions. The result should be treated as indicative only.

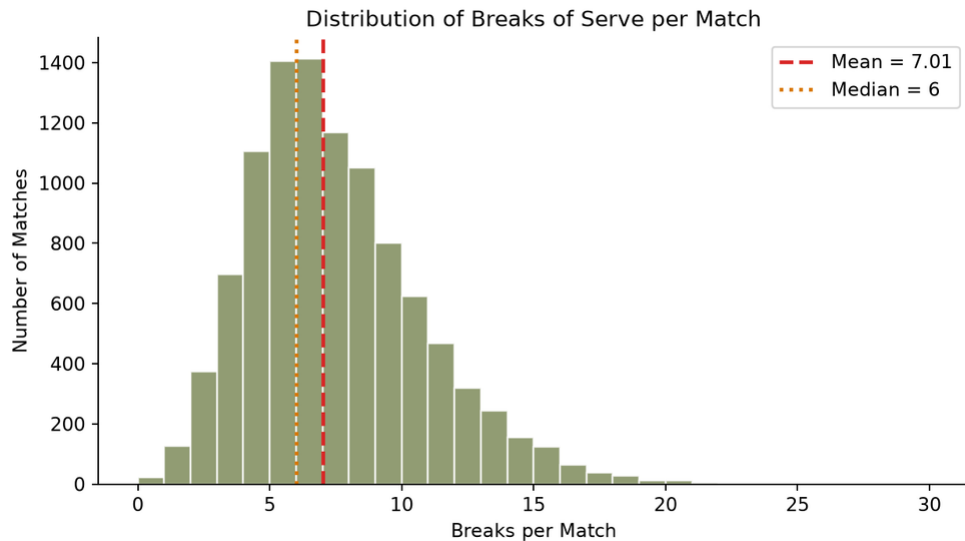


FIG. 13 — DISTRIBUTION OF BREAKS OF SERVE PER MATCH

What Is the Average Number of Breaks of Breaks of Serve per Match?

The mean number of breaks per match is 7.01 and the median is 6. The narrow gap between mean and median indicates a relatively symmetrical distribution with a slight right skew. This right-hand tail is caused by 1.6% of data points in the outlier range — retained as genuine occurrences likely from long matches with poor serving quality or tight competition between two strong returners.

7.01

Mean breaks per match · Median: 6

1.6% outliers retained as genuine. 36 matches with only 1-2 games were validated against score data and confirmed as short matches, not incomplete data — so they were not excluded.

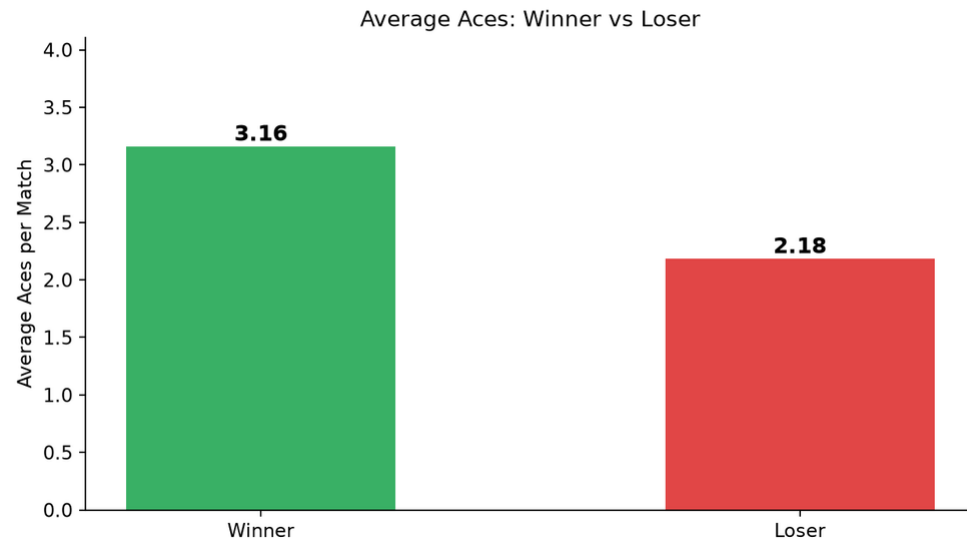


FIG. 14 — ACES PER MATCH: WINNER VS LOSER

Does the Winner Usually Play More Aces?

Winners average 3.16 aces — approximately 45% higher than the 2.18 average for losers. However, this pattern does not hold in every match: winners recorded more aces in only 49.9% of matches, while ace counts were tied in 23%. In the remaining 27%, the loser actually hit more aces but still lost. Aces alone are not a definitive predictor of outcome — it's a general statistical advantage, not an absolute rule.

3.16

Winners · avg aces

2.18

Losers · avg aces

49.9%Matches where
winner had more
aces

Matches with missing winner_code were excluded. The 45% average gap coexists with only 49.9% match-level consistency — a classic case where group averages mask individual-match variability.

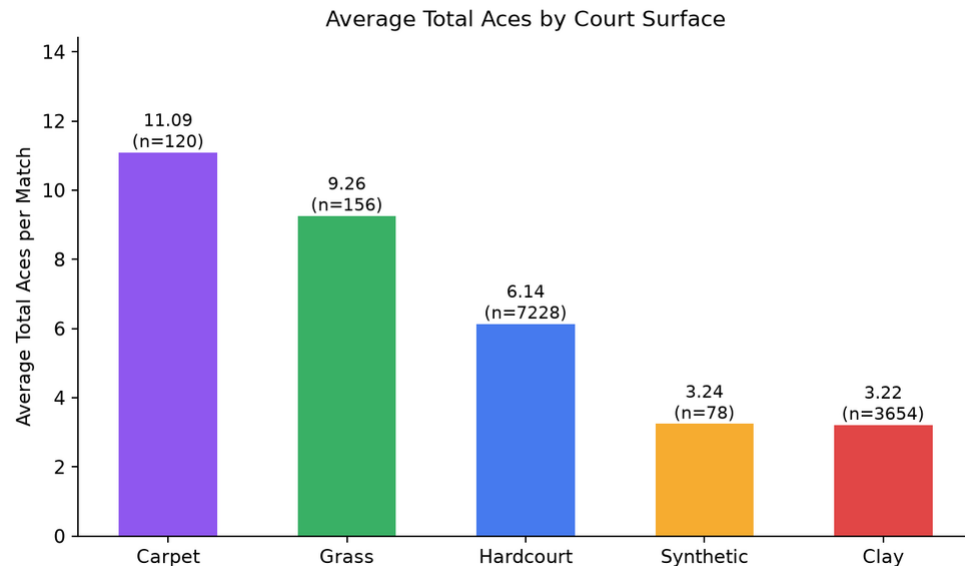


FIG. 15 — AVERAGE ACES PER MATCH BY COURT SURFACE

Does the Court Surface Affect the Number of Aces?

and clay yield the lowest. Fast courts produce significantly more aces than slow courts — a difference of nearly 3.5× between the highest and lowest rates. This aligns perfectly with tennis physics: on clay, the ball moves more slowly and bounces higher, making service returns easier; on grass and carpet, the ball maintains speed, increasing the likelihood of aces. Grass and carpet yield the highest ace rates; synthetic

Grass · high

Carpet · high

Hardcourt · mid

Synthetic · low

Clay · lowest

3.5×

Gap between fastest & slowest surfaces

Fragmented surface categories (green clay, indoor red clay) were consolidated into five main groups: Hardcourt, Clay, Grass, Carpet, and Synthetic — ensuring statistical significance through adequate sample sizes.

SECTION THREE / THREE

03

Match Dynamics & Data Quality

PRESENTED BY Yasamn Aghaei

QUESTIONS IN THIS SECTION

Q3 Highest number of wins

Q5 Typical number of sets

Q9 Most tournaments won by a player in a single month

Q14 Most common surface

OP2 Same players: Hard vs Clay

Q4 Longest match by duration

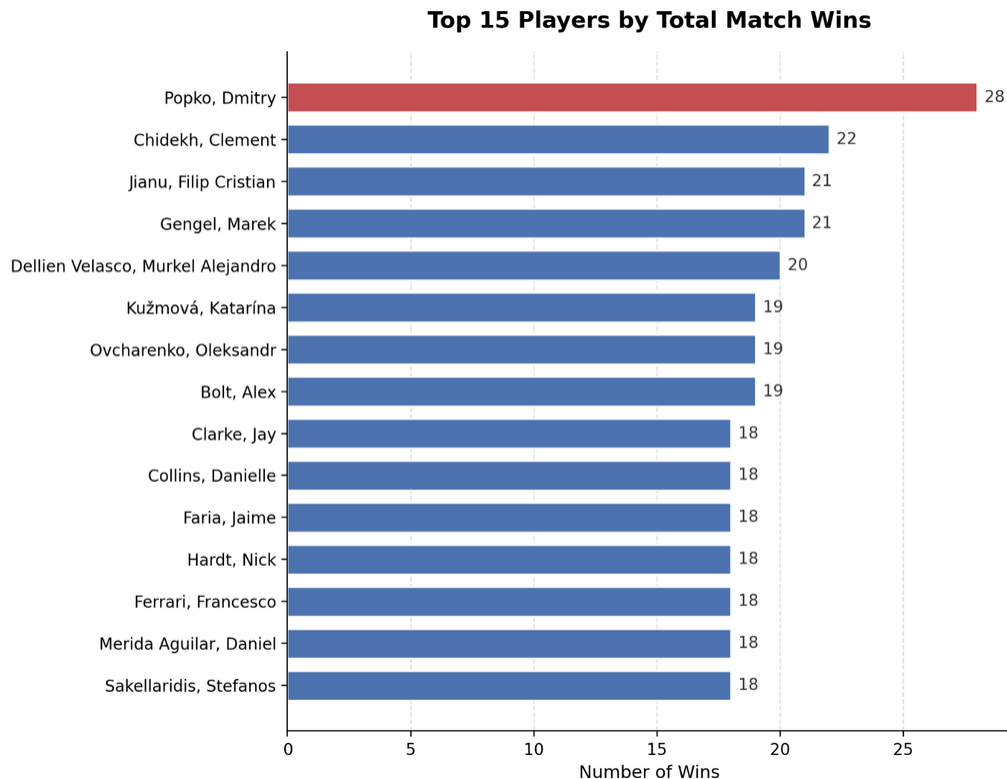
Q6 Most successful countries

Q11 Average match duration

OP1 Duration by surface

★ Scoring-rule validator

Fig. 16 — TOP 15 PLAYERS BY TOTAL MATCH WINS



Which Player Has the Highest Number of Number of Wins?

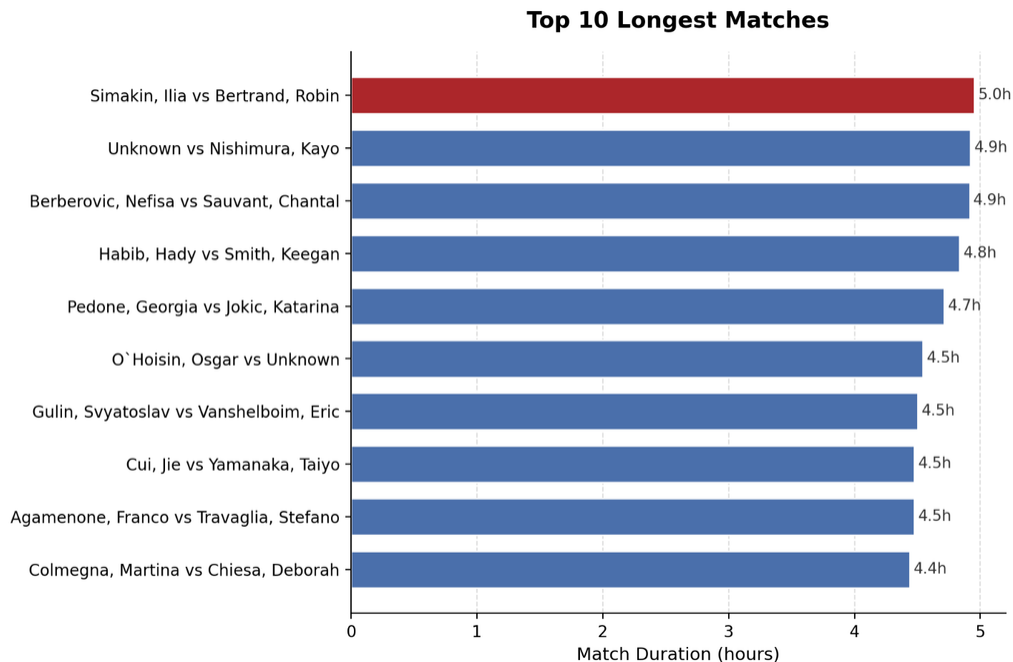
Popko leads with **28 wins**, followed by Chidekh (22) and Jianu / Gengel (21 each). The ranking rests on 11,674 wins linked to a player row — about 81% of completed matches with a recorded winner. This is a volume ranking, not a quality ranking: total wins mix how often a player competed and how often they won, so ITF-heavy schedules can produce higher totals than fewer, harder matches at higher tiers.

28

POPKO — HIGHEST TOTAL WINS

2,739 wins were dropped because the winning side had no player row in home_team/away_team. Completed matches with non-null winner_code and score_ready; wins attributed via winner_code (1=home, 2=away).

Fig. 17 — TOP 10 LONGEST MATCHS



What Is the Longest Match by Duration?

The longest match is **Simakin vs Bertrand** (Monastir, ITF Men, hard outdoor), lasting 4 hours 57 minutes over 35 games on 2024-02-10. The top 10 matches sit between roughly 4.4h and 5.0h. The 300-minute upper cap means this is "longest inside the plausible range," not a claim that no real match exceeded five hours — documented long matches (Federer vs del Potro, 4h 26m at London 2012 Olympics) stay below this threshold.

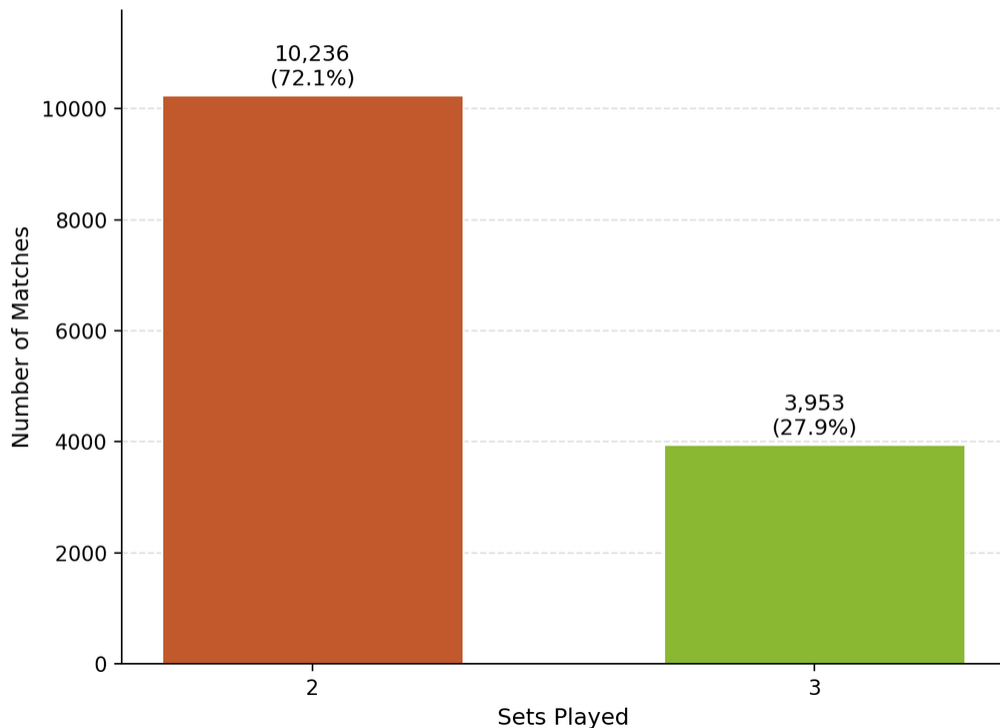
4h 57m

SIMAKIN VS BERTRAND — 35 GAMES

Duration pipeline: duration_reliable + 15–300 minute window + completed matches + games cross-check. Lower bound (15 min) removes 3 retirement-like rows; upper bound (300 min) removes 14 suspect extreme values. Same pipeline used for Q11 and surface-duration analyses.

Fig. 18 — Typical number of sets played

Typical number of sets Played in a Tennis Match = 2



What Is the Typical Number of Sets per per Match?

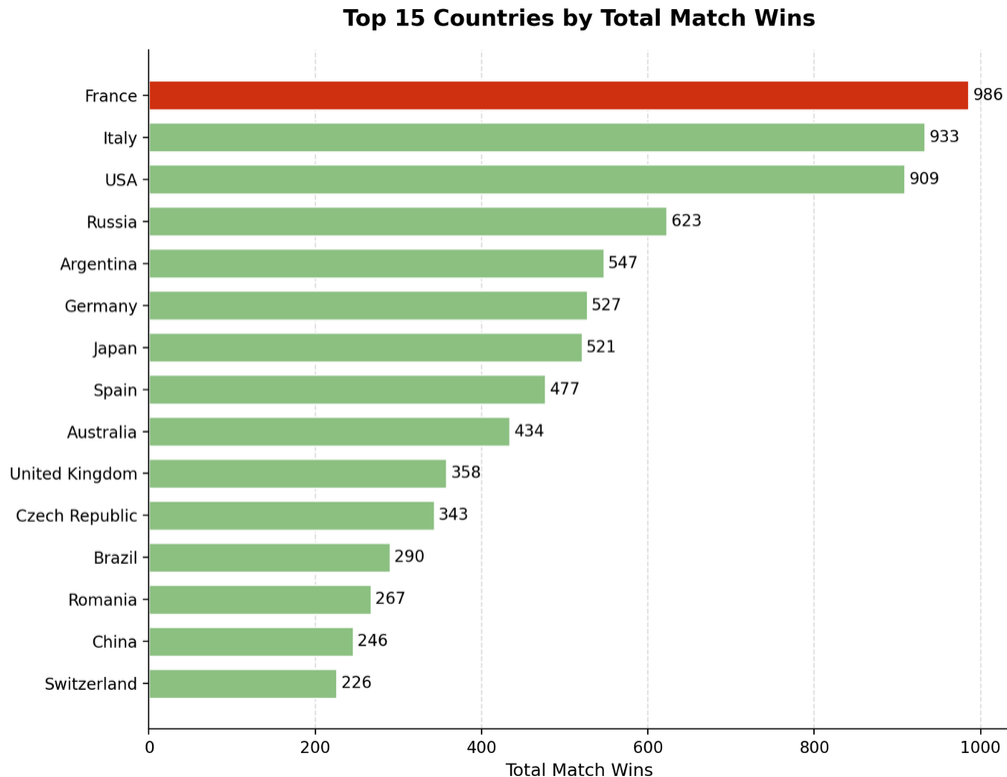
Of 14,189 completed matches, **72.1% end in 2 sets** and 27.9% in 3 sets (mean 2.28). This matches the best-of-3 structure perfectly: straight-set wins stop at two sets, so a ~72/28 split means most matches do not reach a decider. Agreement with the independent power table is 100% on 10,074 overlapping matches, confirming the result is robust.

72.1%

MATCHES ENDING IN 2 SETS

sets_played aligned from home and away score tables (max of the two); restricted to match_completed; cross-checked against max set_num from power table. 100% agreement on 10,074 overlapping matches validates the alignment logic.

Fig. 19 — TOP 15 Countries by Total Match Wins



Which Country Produces the Most Successful Successful Players?

By total wins: France (986), Italy (933), USA (909). But by wins per player (with player (with ≥ 50 players): Argentina (5.58), Brazil (5.00), France (4.88), Spain Spain (4.87), Czech Republic (4.83). "Most successful" is not a single fact — total fact — total wins reward large player pools, while wins per player rewards rewards efficiency but is sensitive to the cutoff threshold.

France 986

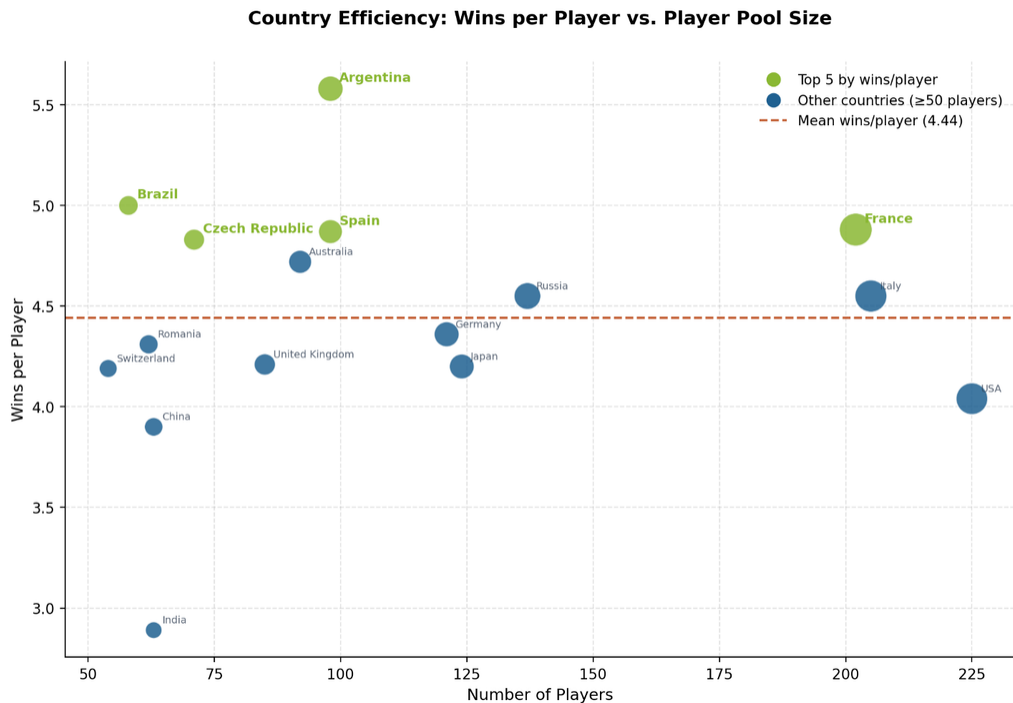
TOTAL WINS LEADER

Argentina 5.58

WINS / PLAYER LEADER

Same win attribution as Q3, grouped by country. Secondary metric = total wins / distinct players with a minimum player-count threshold. A few players with multiple country labels were left as diagnostic noise rather than force-merged.

Fig. 20 — Wins per Players vs. Player Pool Size



Which Country Produces the Most Successful Successful Players?

THE BETTER QUESTION IS:

Which Countries Are Most Efficient by Wins Per Player?

Among countries with ≥50 players, Argentina leads efficiency at 5.58 wins/player, followed by Brazil (5.00), France (4.88), Spain (4.87), and Czech Republic (4.83). The 50-player threshold filters out small samples. Efficiency differs from raw win totals — high-volume countries do not always dominate per-player output.

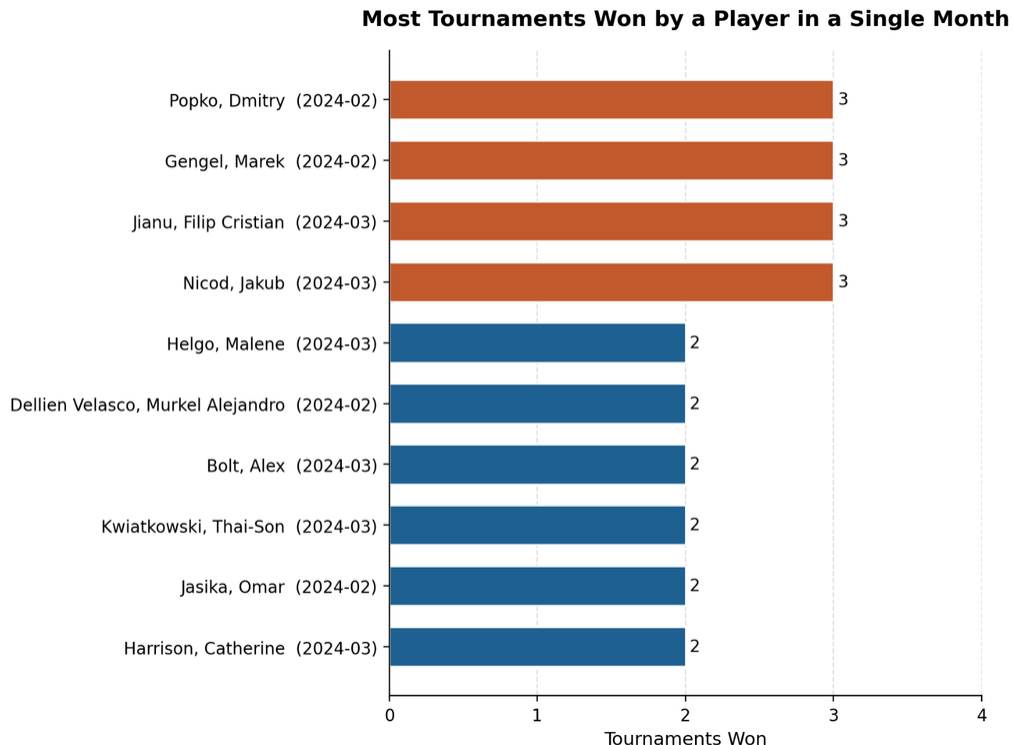
Brazil
5.00

WINS / PLAYER LEADER

Argentina
5.58

WINS / PLAYER LEADER

Fig. 21 — Most Tournaments Won by a Player in a single Month



What Is the Most Tournaments Won in a Single Single Month?

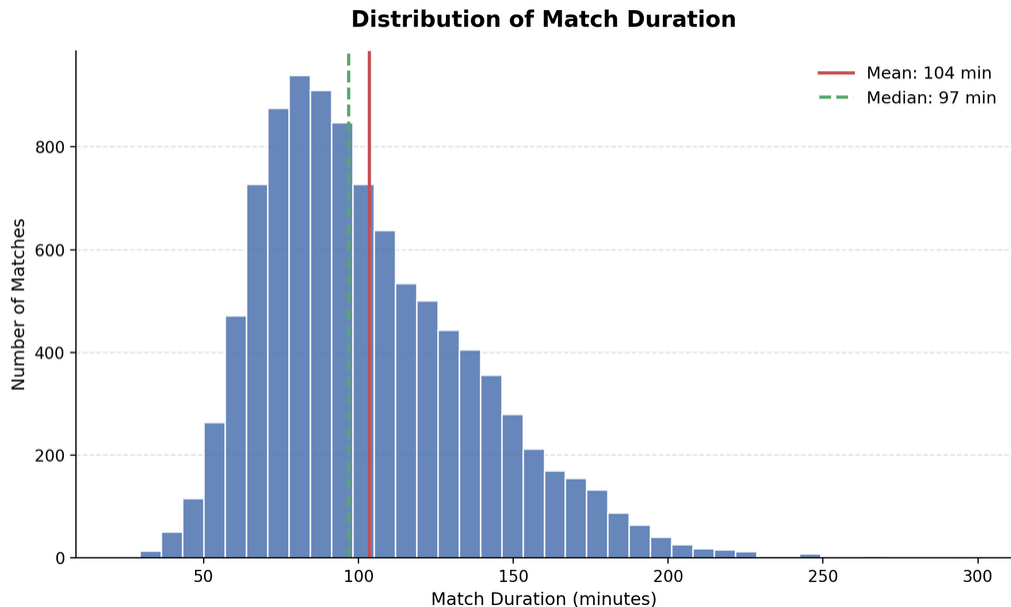
The maximum is **3 tournaments in one month**, reached by four players — Popko, Gengel, Jianu, and Nicod — in Feb–Mar 2024, all in ITF Men. Three events in one month fits the weekly ITF calendar structure. The same pattern is less likely at ATP level because events are spaced further apart and fields are deeper. Of 260 main-draw finals, 28 had no recorded winner and were excluded.

3 tournaments

IN ONE MONTH — 4 PLAYERS (FEB–MAR 2024)

Restricted to main draw (qualifying excluded); is_final matches only; required completed match with winner; player attributed as in Q3; grouped by player × month. Measured only over months present in the data.

Fig. 22 — Distribution of Match Duration



What Is the Average Match Duration?

Using the same duration pipeline as Q4: mean **103.6 minutes**, median 96.8, standard deviation 35.2 (n=10,075). The mean sitting above the median indicates right-skew. By sets: ~87.8 minutes for 2-set matches vs ~139.6 minutes for 3-set matches — the ~52-minute gap is consistent with the cost of one extra full set.

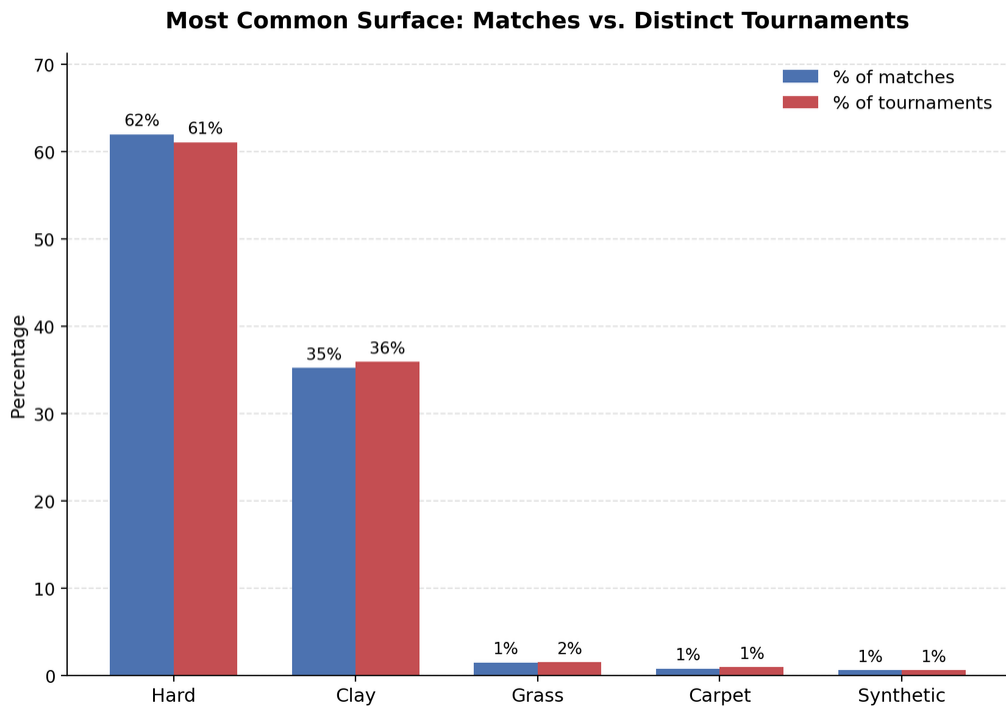
103.6 min
MEAN (N=10,075)

96.8 min
MEDIAN

87.8 →
139.6
2 SETS → 3 SETS
(+52 MIN)

duration_reliable + 15–300 minutes + completed matches + games cross-check. Same pipeline as Q4. Describes the filtered population only — retirements and corrupted/incomplete durations are out of scope by design.

Fig. 23 — Most Common surface



What Is the Most Common Surface?

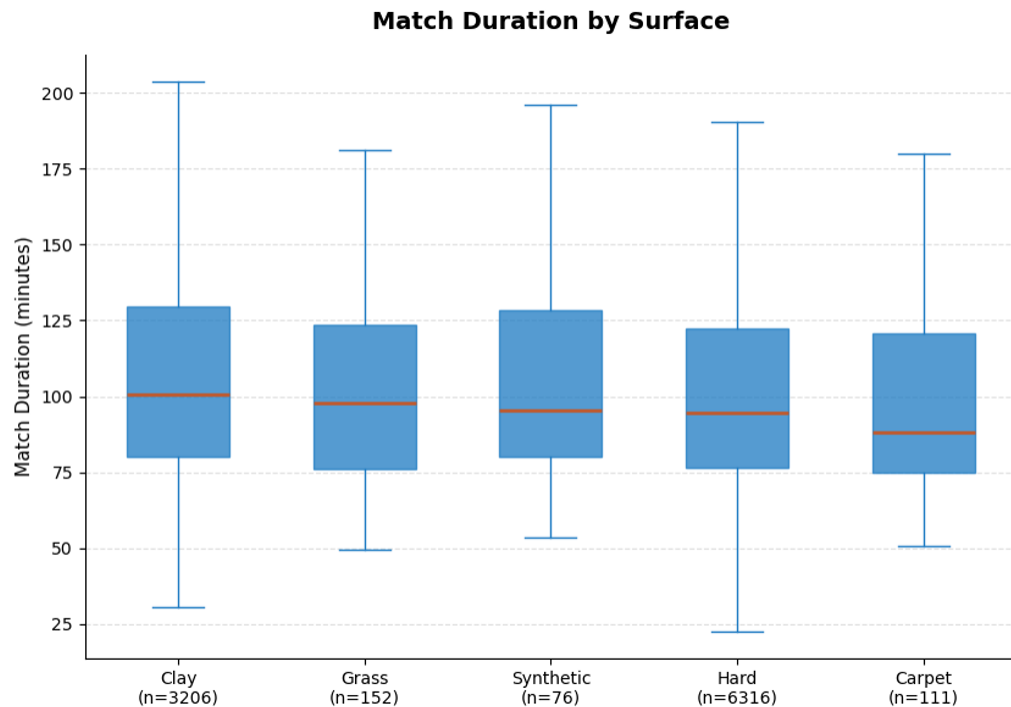
Hard court is first at both levels — ~62% of matches and ~61% of tournaments — with Clay second at ~35–36%. Match-level and tournament-level shares are close, so draw-size differences by surface are small in this dataset. 8 of 529 tournament groups had mixed surface labels (mode used for resolution); the inconsistency is real data noise, not silently fixed.

~62%

HARD COURT — MOST COMMON SURFACE

Match-level counts from `ground_type`; tournament-level via mode of surface within `(tournament_name, category)`. Broad families: Hard / Clay / Grass / Carpet / Synthetic. 8 mixed-label groups resolved via mode — ranking unchanged.

Fig. 24 — Match Duration by Surface



How Does Match Duration Vary by Surface?

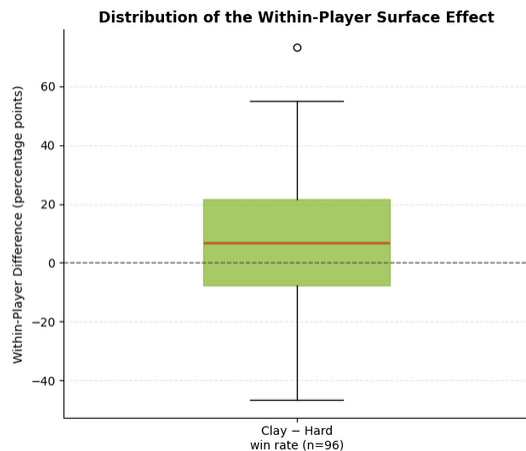
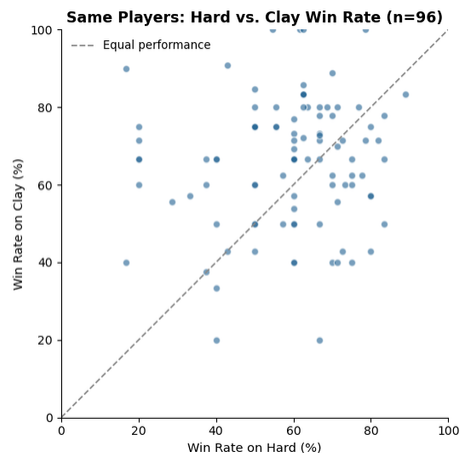
Median match minutes by surface: Clay **100.8** (n=3,206), Grass 98.0 (n=152), Synthetic 95.4 (n=76), Hard 94.6 (n=6,316), Carpet 88.2 (n=111). The only large-sample contrast is Clay vs Hard — about 6 minutes at the median. Grass, Synthetic, and Carpet medians rest on much smaller samples and are less stable.

100.8 min

CLAY MEDIAN (N=3,206) — LONGEST

Same duration pipeline: duration_reliable + 15–300 min. Outliers hidden on the boxplot for readability. Only Clay (n=3,206) and Hard (n=6,316) have sample sizes large enough for stable conclusions.

Fig. 25 — Players Win Rate On Hard vs. Clay



Do the Same Players Perform Differently on on Hard vs Clay?

Among 96 players with ≥ 5 matches on both surfaces, the mean within-player difference (Clay - Hard) is **+6.8 percentage points**: 55 players were better on Clay, 36 on Hard, 5 tied. Paired t-test: $t=2.94$, $p=0.004$. Each player is their own control — the test asks whether the average of (Clay rate - Hard rate) differs from zero. p is not effect size; +6.8 pp is modest.

+6.8 pp

CLAY - HARD ($T=2.94$, $P=0.004$)

≥ 5 -match threshold on both surfaces. Dependence across a player's matches is not modelled — the t-test is a first-pass average test, not a full causal model. Individual differences spread widely, so the mean does not describe most players.

Tennis Scoring-Rule Validator — Data Quality Quality Layer

A scoring-rule validator aligns sets_played between home and away score tables, classifies every set under correct tennis rules (6-x / 7-5 / 7-6, match/super TB, advantage sets), and flags genuine anomalies — illegal 7-x scores, tiebreak mismatches, winner-vs-games contradictions, best-of-3 overruns, and implausible durations. This validator is the upstream filter that makes every Q3–Q14 denominator defensible.

status	condition	meaning
valid_normal	6–0 ... 6–4, or 7–5	Standard completed set
valid_tiebreak	7–6	Set decided by a set tiebreak
valid_super_tb	≥10 games with margin ≥2, and the set is the deciding set	Match / super tiebreak
valid_advantage	8–6, 9–7, ...	Advantage set without a set tiebreak (special formats)
anom_7	7–0 through 7–4	Impossible under standard scoring rules
incomplete_last_set	Does not reach a valid endpoint and is the last set played	Retirement / default pattern
genuine_anomaly	Does not reach a valid endpoint and the match continued past it	True scoring inconsistency
anomaly_one_sided_null	Only one side has a score recorded	Incomplete recording

~1,490

MATCHES WITH BAD SET STATUS

~1,721

ANOM_7 SET-LEVEL CASES

14,413

SCORE_READY (USABLE)

275

RETIREMENT-LIKE MATCHES